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MILITARY SPECIFICATION

MILLING MACHINES, VERTICAL SINGLE AND MULTIPLE SPINDLE, COMPUTER NUMERICALLY CONTROLLED (CNC) AND TRACER CONTROLLED PROFILER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three-axis computer numerically controlled (CNC) and tracer controlled vertical spindle(s) profiler. The spindle(s) may be mounted stationary with Y Axis traversing under the heads or the spindle(s) may be mounted on a ram or column that traverses the heads across the work table.

1.2 Classification. The milling machines covered by this specification shall be of the following types, classes, and sizes and shall comply with the applicable minimum requirements set forth in table I. The types, classes, and sizes shall be as specified (see 6.2.1).

Type I - Single-spindle machine, single-drive motor.

Class 1 - One spindle.

Type II - Multi-spindle machine, two or more spindles in a single unit driven by one motor.

Class 1 - Applicable to type I machines only.
Class 2 - Two spindles, specify distance between spindle centers (see 6.2.1).
Class 3 - Three spindles, specify distance between spindle centers (see 6.2.1).
Class 4 - Three spindles, specify distance between spindle centers (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - Multi-spindle machine, two or more spindles, each as an individual unit and independently driven.

Class 1 - Applicable to type I machines only.
Class 2 - Two spindles, specify distance between spindle centers (see 6.2.1).
Class 3 - Three spindles, specify distance between spindle centers (see 6.2.1).
Class 4 - Three spindles, specify distance between spindle centers (see 6.2.1).
Class 5 - Four spindles, specify distance between spindle centers (see 6.2.1).
Class 6 - Four spindles, specify distance between spindle centers (6.2.1).

Size	Table work surface (inches)
1	28 by 72
2	28 by 119
3	28 by 143
4	40 by 72
5	40 by 119
6	40 by 143
7	30 by 119
8	40 by 119
9	40 by 143

Type I, class 1, includes all of the above sizes. Type II, classes 2 through 4, includes sizes 1 through 6. Type III, classes 2 through 6, include sizes 7 through 9.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

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FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety Health Standards (29 CFR 1910)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360.02 - Manual For Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

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ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B11.8 - Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care and Use.

(Application for copies should be addressed to the American National Standards Institute, Department 969, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, Sales and Service Department, 1916 Race Street, Philadelphia, PA 19103.)

ELECTRONICS INDUSTRIAL ASSOCIATION (EIA)

RS-227 - One Inch Perforated Tape.

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-244 - Character Code for Numerical Machine Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/ Positioning Numerically Controlled Machines.

RS-281 - Electrical Construction Standards - Machine Numerical Control.

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

RS-494 - 32 Bit Binary CL (BCL) Exchange Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to Electronics Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization For Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

NATIONAL AEROSPACE INDUSTRIES ASSOCIATION STANDARDS (NAS)

NAS 979 - Uniform Cutting Tests

(Application for copies of National Aerospace Standards should be addressed to Aerospace Industries Association, 1725 DeSales Street, N.W., Washington, DC 20036.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheet or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be one of the manufacturer's current models, new and unused, equipped with a three-axis contouring computer numerical control (CNC) of a solid-state design. The control unit shall be capable of receiving instructions from manual data input, punched tape, flexible diskette, or direct computer input, and automatically command the machine to perform milling and profiling operations by controlling the movements along the X, Y and Z axes individually, simultaneously, or in any

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combination at all programmable feed rates. The axes of movements shall be in accordance with Electronic Industries Association Standard RS-267: (1) X axis - longitudinal movement of table, (2) Y axis - movement of the ram or column over and across the table or movement of the table under the spindles(s), (3) Z axis - vertical movement of spindle head mounted on end of ram extending over the table. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The milling machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the milling machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components,

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mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 and ANSI B11.8 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible. In the event of a lube failure, a means shall be provided to automatically inhibit axis movement until the problem is corrected.

3.2.9 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or the machine.

3.3 Construction. The milling machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solder or paste shall not be used on castings and forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, and soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified

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herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standards" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameter shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.4 Components. The machine shall consist essentially of a fixed height bed, table, cross slide (ram), spindle carrier, and a continuous path computer numerical control system.

3.4.1 Bed. The bed shall be of fixed height, ribbed and cross ribbed for strength and rigidity, constructed to dimensions to ensure proper alignment and vibration-free operation performance while supporting components mounted thereon and therein. Access doors or opening in the bed shall be provided for servicing motors and other components used therein. The bed shall be integrally made with the column or shall be separately made and machined to be rigidly attached to the column with the ways of each component in proper alignment as stated in table II. The bed ways shall be properly spaced and be of suitable size and hardness for long life under heavy loads. An adequate number of leveling screws or hold down bolts shall be provided to enable the machine to be properly installed and leveled.

3.4.2 Table. The table shall be adequately proportioned to provide sufficient rigidity for acceptable performance throughout the full range and capacity of the machine. It shall be accurately machined and fitted to the bed ways with means for adjustments to require alignment accuracies as

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specified in table II. The table shall be provided with T-slots machined in accordance with ANSI B5.1, table I and parallel to the longitudinal travel of the table. Channels and catch basins for chips and coolant with one or more openings, with provisions to prevent chips from entering, shall be provided within the perimeter of the table for returning of coolant to the sump. Means shall be provided to prevent overtravel of the table.

3.4.3 Cross slide base. The cross slide base shall be of adequate proportion, ribbed and braced for strength and rigidity to support the cross slide. The base shall be machined to be rigidly attached to the bed with the ways of each component in proper alignment. The cross slide base ways shall be of size and finish to assure low friction and long life while supporting the cross slide in its movements. Access doors or openings shall be provided for servicing components housed therein. An adequate number of hold-down bolts and leveling screws shall be furnished to enable the machine to be properly installed.

3.4.4 Cross slide. The cross sliding member shall be constructed for strength and rigidity, with accurately machined and finished ways. It shall provide vibration-free performance, minimum sag and vertical ways square with the table for spindle head travel. The cross slide movement shall be parallel to the table top within the limits specified in table II.

3.4.5 Spindle carrier. The spindle carrier shall be a vertical sliding headstock adequately proportioned to capably perform milling operations within full range and capacity of the machine. The carrier shall have precision ways for long life and shall be fitted to the vertical ways of the cross slide with gibs to properly adjust and align the components. The spindle carrier shall incorporate a counterbalance device adequately proportioned to support the weight of the carrier and spindle assembly.

3.4.6 Spindle. The spindle(s) shall be made of high quality alloy steel, heat treated, hardened to Rockwell C-45 $\pm$3 and ground to a precision finish of not greater than 18 microinch. It shall be supported by a sufficient number of precision bearings to provide axial and radial support to maintain requirements set forth in table II. The spindle shall be designed and assembled in such a manner that the effect of thermal expansion on accuracy is held to a minimum. The spindle shall have a precision ground National Standard Taper in accordance with table I. Except when power drawbar is provided, the spindle shall be provided with an arrangement for quick changing arbors and adapters. Means shall be furnished to start, stop, jog, reverse, select speeds and lock the spindle. The number of spindle speeds and range of speeds shall be as specified in table I. When multiple spindles are required, they shall be in line with the X axis and perpendicular to the table work surface.

3.4.7 Axis drives. Movement along the X, Y, and Z axes shall be actuated through a closed-loop, high-gain electric servo mechanism driving antibacklash ball screws with double preload recirculating nuts.

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Infinitely variable feed rates shall be provided for all axes actuated and controlled by commands from the control tape, manual data input, or manual jog control.

3.4.8 Power rapid traverse. Power rapid traverse may be independent of any feed adjustment and shall be provided to each axis, X, Y and Z.

3.4.9 Way wipers and covers. Wipers and covers or other means shall be provided to protect the way bearing surfaces from dust and other foreign matter that may be harmful to the finish.

3.4.10 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.11 Hydraulic system. The hydraulic system when furnished, shall be complete, including all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.12.1 Motors. Motors shall be rated continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operated on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall be in accordance with ANSI/NEMA MG 1.

3.4.13 Solid-state components. Electrical devices used to power electrical servo and spindle motors or control voltage delivered to solid-state electrical components shall be solid-state devices. Such devices shall be capable of handling the electrical load imposed by operation of the machine at its maximum capability. The use of selenium in solid-state components shall be restricted to devices that protect other solid-state components from voltage surge.

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3.4.14 Computer numerical control (CNC). Unless otherwise specified (see 6.2.1), the milling machine will be equipped with a fully automatic solid-state soft-wired, integrated circuit type numerical control. The numerical control shall provide automatic control of machine functions, operating modes, spindle speeds, direction of rotation and simultaneous control of the machines three-linear axes in linear and helical interpolation. The control shall also be provided with circular interpolation capabilities. Machine axes shall be identified in accordance with EIA RS-267. Line voltage fluctuation of ±10 percent from the normal operating voltage shall not adversely affect the operation of the CNC. Unless otherwise specified (see 6.2.1), the control shall be provided for connection to a 115-volt alternating-current (AC), 60-Hz, single-phase power source. The control shall be capable of operating in an outside ambient temperature of 50 to 110 degrees Farenheit (F) at 5 to 95 percent noncondensating relative humidity. The control shall automatically shut down when the maximum operating temperature is reached. Program resolution shall be 0.0001 inch or 0.001 millimeter (mm) for each linear axis. The mechanical and electrical design of the control shall comply with EIA RS-281 construction standards. The control shall be capable of accepting input data in the form and format specified by the ordering activity.

3.4.14.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be 1-inch, eight-channel punched tape complying with EIA-RS-227.

3.4.14.1.1 Tape reader. When 1-inch tape is the specified input media, the control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust-proof enclosure mounted in a control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.14.2 Flexible diskette data input. When 32 Bit Binary Cutter Location (BCL) Exchange Input for numerically controlled machines is specified (see 6.2.1), the input media shall be an 8-inch flexible diskette recorded in accordance with 3.4.14.2.1. The input data format shall comply with EIA RS-494. Manufacturers providing functions or features that are not

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currently covered in EIA RS-494 shall apply to the Electronic Industries Association, EI-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the manufacturer's features or functions. The control shall be equipped with an 8-inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundance check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.14.2.1 Flexible diskette format. When an 8-inch flexible disk is specified as the input media to the numerical control, it shall be a single density recording on one side using a soft-sectored format. The disk shall be divided into 77 distinct locations on the single-side disk. Tracks shall be assigned numbers for 00 through 76. Track 00 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 00, 74, 75 and 76 shall not be used. There is an index hole in the jacket and the disk that shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket exposing the disk at the read/write head slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read/write head. Each track shall be divided into 26-addressable sectors which contain 128 bits. All sectors shall be identified by the file starting sector number with track 00 sector 1, designated as 0. The first part program shall be located at sector 1 E hexidecimal corresponding to track 1, sector 5. Where 2-byte numbers are referenced, the high order byte shall be recorded first and the low order last. Track 1, sectors 1 through 4 shall contain the program director of up to 16-part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry. Unused directory entries shall point to the next available sector after the last file and contain the 30-character ASCII space characters. The last sector of each file shall be padded with end of the file (EOF) records (8000 0000). Diskette addressing shall comply with the requirements of table III.

3.4.14.3 Program features.

3.4.14.3.1 Tape coding. The control shall accept punched tape programmed in accordance with EIA RS-274. The control shall be capable of recognizing and automatically accepting coding in accordance with EIA RS-244 or EIA RS-358.

3.4.14.3.2 Leading zero suppression. The control shall be capable of suppressing leading zeros.

3.4.14.3.3 Selectable inch/metric input The control system shall provide operator or program selectable inch/metric input capability.

3.4.14.3.4 Absolute/incremental input. The control shall have switchable absolute/incremental input capability.

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3.4.14.3.5 Departure control. The minimum departure permitted by a single command in either incremental or absolute model shall be not less than plus or minus 99.9999 inches or 999.999 mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.14.3.6 Preparatory function (G-codes). The preparatory function codes of the control shall be in accordance with EIA RS-274.

3.4.14.3.7 Miscellaneous functions (M-codes). The miscellaneous function codes of the control shall be in accordance with EIA RS-274.

3.4.14.3.8 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Inches per minute (ipm)/millimeter per minute (mmpm).
d. Vector feedrate.

3.4.14.4 Axis and tool management.

3.4.14.4.1 Tool length offsets. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset feature shall have a range of not less than the following:

a. Inch system ±0.0001 inch to ±1.0000 inch.
b. Metric system ±0.001 mm to ±25.400 mm.

3.4.14.4.2 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system ±0.0001 inch to ±1.0000 inch.
b. Metric system ±0.001 mm to ±25.400 mm.

3.4.14.4.3 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.14.4.4 Reversal error compensation. Reversal error compensation circuitry shall be provided for a selectable dimension correction to be added automatically to the command dimension to each linear axis reversal to compensate for mechanical lost motion.

3.4.14.4.5 Lead error compensation. Lead error compensation software shall be provided to permit compensating for the lead errors of each axis ball screw.

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3.4.14.4.6 Auxiliary functions. Unless otherwise specified (see 6.2.1), the CNC shall respond to the following program command auxiliary functions:

a. Interpolation.
b. Fixed cycles.
c. Dwell.

Unless otherwise specified (see 6.2.1), miscellaneous function shall include program stop, end of program, spindle clockwise (cw), spindle counterclockwise (ccw), spindle off, coolant on-off tool change, end of program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274.

3.4.14.5 Operation interface controls.

3.4.14.5.1. Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis (may be a LED display on the control unit).
d. Control and system error messages.

3.4.14.5.2 Control system modes of operation. The operational modes of the CNC control shall include manual, automatic, single-block, MDI, and sequence number search.

3.4.14.5.3 Manual data input. A MDI unit shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.14.5.4 Block delete. The control shall have a block-delete feature to permit bypassing of blocks of programmed data.

3.4.14.5.5 Block-by-block read. The control shall have block-by-block read function to provide for reading information one block at a time.

3.4.14.5.6 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.14.5.7 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

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3.4.14.5.8 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from 0 to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent.

3.4.14.5.9 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 and 120 percent of programmed speeds.

3.4.14.5.10 Mirror image. The control shall be capable of producing either right- or left-hand parts from the same input data through the inversion of X and Y axis. Inversion of each axis shall be by keyboard by switch or by programmed entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.4.14.5.11 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus direction.

3.4.14.5.12 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.14.5.13 Program tryout. The control shall have a program tryout feature to permit new part programs to be read through the control without machine motions.

3.4.14.5.14 Operator control panel. The control unit shall have a control panel which provides, unless otherwise specified (see 6.2.1), at least the following control functions:

a. Emergency stop.
b. Cycle start.
c. Optional stop.
d. Mode selection.
e. Feed hold.
f. Feedrate override.
g. Spindle-speed override.
h. Jog.
i. Jog direction.
j. EOF stop.

3.4.14.6 Miscellaneous control requirements.

3.4.14.6.1 Part program storage. Part program memory loadable from the data input media shall be provided. Unless otherwise specified (see 6.2.1), the part program memory shall be capable of storing not less than 100 feet of postprocessed part program tape.

3.4.14.6.2 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory, to hold the size memory provided at full load for a period of not less than 72 hours when the primary power source fails.

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3.4.14.6.3 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator's actions.

3.4.14.6.4 Position feedback device. A position feedback device including all related components, hardware and electrical equipment shall be provided for each axis having a readout system. The device shall provide axis travel feedback over the entire range of axis travel for each numerically controlled axis. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device during operation of the machine. Operation of the position feedback device shall permit the machine to perform within the tolerances and accuracies given in table II.

3.4.14.6.5 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided as part of the executive program or as a separate program.

3.4.14.6.6 Maintenance diagnostic systems. Maintenance diagnostics software shall be furnished in either the control as nonvolatile memory or in the form of 1-inch tape or 8-inch flexible disk, as required by the specified input media, to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.14.6.7 Peripheral equipment interface. The control shall be equipped with an interface complying with EIA RS-232.

3.4.14.6.8 Canned cycles. Unless otherwise specified (see 6.2.1), the following canned cycles shall be provided.

a. Drilling.
b. Bolt hole circle.

Such cycles shall be called by G codes and shall be initiated when X, Y, Z, and R plane data is entered plus data for the F, E, and K words.

3.4.14.6.9 Operator portable or pendant control. When specified (see 6.2.1), an operator's control of the pendant or portable box type shall be provided for the CNC with features as specified by the procuring activity.

3.4.14.6.10 Postprocessor. Unless otherwise specified (see 6.2.1), the CNC shall be furnished with postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified part program language (see 6.2.1), and shall be written for use with the computer specified (see 6.2.1). Documentation sufficient to implement and use the postprocessor

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shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the milling machine.

3.4.14.6.11 Additional data input media. When specified, a secondary input device shall be provided to input data on either 1-inch, eight-channel punched tape, magnetic tape cassette or flexible diskette (see 6.2.1). When 1-inch, eight-channel punched tape reader is specified, the tape reader and control shall satisfy the requirements in 3.4.14.1.1.

3.4.14.6.12 Reader/recorder. When specified (see 6.2.1), the control shall be equipped with a reader/recorder for reading and recording either tape cassettes or flexible disks. When specified (see 6.2.1), a portable tape punch shall be supplied for punching 1-inch, eight-channel tape. Recording shall be accomplished when the control is operated in the program record mode.

3.4.14.6.13 Tracer machining capability. When specified by the procuring activity (see 6.2.1), an integrated tracer function shall be incorporated into the CNC unit. Its function shall be to provide three-dimensional duplication of a physical model by responding to the feedback of a tracing stylus as it passes over the model surface.

3.4.14.6.14 Digitizing capability. When specified by the procuring activity (see 6.2.1), a digitizing function for machines equipped with tracing capabilities as described in 3.4.14.6.13 shall be an integral part of the CNC unit. It shall be capable of simultaneously recording the path of the tracer head over a model as the tracer milling function is machining the first of several identical parts. This shall generate coordinate data required to develop a part program thus eliminating the tracer function on additional identical parts.

3.5. Sizes and capacities. The sizes and capacities of the milling machine shall conform to the requirements of table I.

3.6 Performance. The milling machine and numerical control system shall be capable of satisfactory performance when tested in accordance with 4.5 through 4.7.3.4.

3.7 Alignment tolerances. Each milling machine shall meet the alignment accuracies of table II.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.
b. Operational and programming manuals.
c. Spindle load meter.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

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3.10 Additional performance requirements. Performance features required in addition to or in lieu of the requirements herein shall be as specified (see 6.2.1).

3.11 Marking of instruction plates, indicating plates and nameplates. All words on instruction plates, indicating plates and nameplates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

3.11.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), lubrication chart or plate shall be permanently and securely attached to each milling machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each milling machine. The nameplate shall contain the information listed below. If the milling machine is a special model, the model designation shall include the mode of the basic standard mill and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's serial number
Power input (volts, total amps, phase frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacturer

3.12 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative

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hours, the meter readout shall automatically revert to zero and continue to totalize time.

3.13 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract (see 6.2.2).

3.14 Workmanship. Workmanship of the milling machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1) herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first item submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection

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shall consist of the examination in 4.6 and tests in 4.7 through 4.7.3.4. Failure of the item to pass the first article tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1 through 4.7.3.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and inspection shall be cause for rejection.

4.6 Examination. The milling machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests. Unless otherwise specified (see 6.2.1), the machine and its numerical control system shall be subjected to the following tests. All instruments, tapes, materials and tools required to perform and evaluate these tests shall be furnished by the manufacturer. The measuring instruments shall have evidence of calibration traceable to the National Bureau of Standards. The test tapes and computer printouts shall become the property of the consignee.

4.7.1 Operational tests. The milling machine shall be operated at no load for not less than 30 minutes for an initial warm-up period. Proper operation of all manual controls, motors, adjusting mechanisms and accessories shall be verified during the warm-up period. Upon completion of the initial check procedure, the system shall be cycled continually under tape control to demonstrate operability of the equipment and permit the machine to reach a steady operating temperature. The test tape shall be programmed to initiate a check to insure that the tape reader decoding unit and other systems are properly functioning and present correct machine control information to the machine tool. The program shall also include rapid travel of the slides, independently and simultaneously. A test shall be performed of parity check to demonstrate that the machine shall stop when parity is not obtained due to improper code punching or reading error. If malfunction occurs during the cycling period, the tape shall be restarted each time until a full programmed cycle is completed without a malfunction. Spindle speeds and milling feed rates shall be verified.

4.7.2 Alignment accuracy tests. The milling machine shall be tested to determine compliance with the alignment accuracy requirements of table II. The measuring equipment and machine shall have reached a steady temperature before making any test readings.

4.7.3 Performance tests. The following tests shall be performed to test the functional accuracy, dynamic stiffness, and servo response of the machine and control under operating conditions.

4.7.3.1 Metal removal and maximum horsepower test. The material for this test shall be SAE/AISI 4140 or 4340 annealed steel with Brinell hardness in the range 300 to 350. Not less than 4 workpieces shall be used; thickness not less than 2.5 inches, with area not less than 144 square

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inches. The workpieces shall be placed on the worktable to cover a range requiring not less than 75 percent of the cross and longitudinal travel. The spindle speed and feed rates shall be such that the chip load is not less than 0.012 inch and the peripheral cutter speed shall not exceed 325 surface feet per minute. The depth of cut shall be sufficient to load the spindle motor to its maximum rated horsepower (hp) ± 5 percent. The load meter specified in 3.8.c is acceptable for measuring spindle load, or the manufacturer may use an ammeter and voltmeter with ±2 percent accuracy. Except as limited herein, the cutter speed, feed rate, cutter type, and cutter diameter shall be selected by the manufacturer. The cutting action shall be smooth and even. The surface of the workpieces after the cut shall show no visual evidence of tool chatter, and the surface of the workpieces shall meet the following requirements:

Finish	150 roughness height rating (rhr)
Flatness	0.002 inch per 1-foot span, plus 0.0002 inch per foot accumulative

Metal removal shall be not less than hp by 0.65 cubic inch. With multi-spindle machines, the center spindle only shall be used for the test.

4.7.3.2 Overshoot and undercut test. The overshoot and undercut test shall be performed in accordance with paragraph 4.3.3.4.1 of NAS 979, sheet 19, copyrighted 1969, to test the dynamic servo response of the machine and control. The tolerances for the completed test piece shall not exceed the following:

Overshoot and undercut in X and Y axes	0.0015 inch for Test A
Overshoot and undercut in X and Y axes	0.002 inch for Test B
Overshoot in Z axis	0.0015 inch for Test A
Overshoot in Z axis	0.002 inch for Test B

4.7.3.3 Composite cutting test. The composite cutting test shall be performed in accordance with paragraph 4.3.3.5.1 of NAS 979, sheets 23 and 24, copyrighted 1969, to test the profiling and contouring capabilities of the machine and control unit. A feed rate of 25 ipm and a spindle speed of approximately 1600 revolution per minute (rpm) shall be used for this test. The tolerances for the completed test piece shall be as follows:

Squareness	0.002 inch TIR (outside square, canted square)
Parallelism	0.002 inch TIR (outside square, canted square)
Angularity of ramp	0.0025 inch TIR (5° ramp)
Circle	±0.0025 inch on diameter

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4.3.3.4.1 OVERSHOOT AND UNDERCUT TEST

MACHINE TYPE: ALL PROFILING AND CONTOURING MACHINES.

TEST CRITERIA:

MATERIAL: 7075-T6 ALUMINUM - 12 x 12 x 1.

CUTTER: 1 INCH DIA. - 2 FLUTE HSS END MILL - RIGHT HAND CUT - RIGHT HAND HIGH HELIX - 1 5/8 MAX. LENGTH OF CUTTER TO MINIMIZE CUTTER DEFLECTION

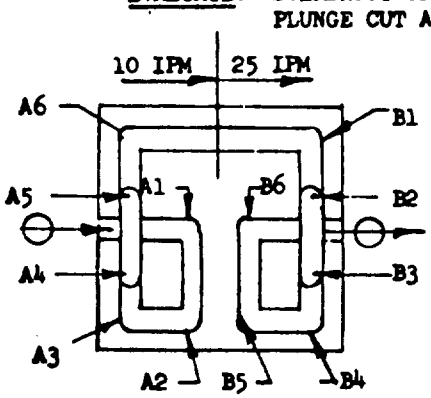
SPINDLE SPEED: MAXIMUM RPM OF MACHINE

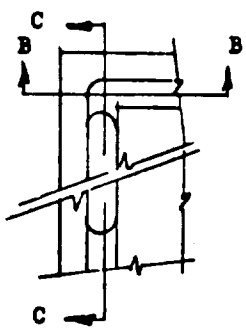
FEED RATE: TEST "A" 10 IPM TEST "B" 25 IPM

SPECIFIED FEED RATE TO BE CONSTANT WITHOUT PROGRAMMED SLOWDOWN. TEST SHALL BE CONDUCTED WITH THE SAME MACHINE CONTROL UNIT SETTINGS AS THOSE OF THE MAXIMUM HORSEPOWER TEST.

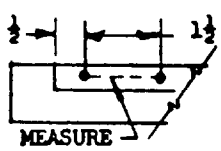
DEPTH OF CUT: .300 X AND Y .300 PLUNGE

EVALUATE: OVERSHOOT OR UNDERCUT ERROR AT EACH 90° CORNER AND EACH PLUNGE CUT AS NOTED.

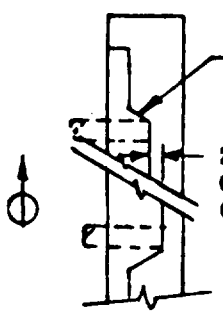




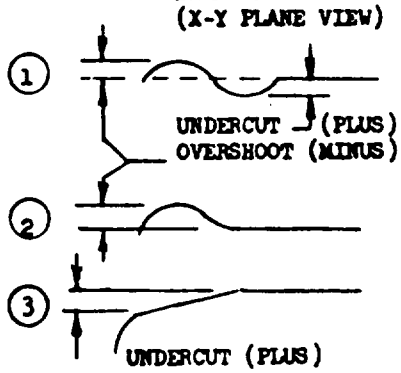
OVERSHOOT OR UNDERCUT 90° CORNER (TYP.) X-Y AXES



OVERSHOOT Z AXIS



EXAMPLES (X-Y PLANE VIEW)



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COMPLETELY REVISED (1)

NAS 979
SHEET 19

APPROVAL DATE APRIL 1956 - REVISION (1) 15 Jan. 1969

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4.3.3.5.1

COMPOSITE CUTTING TEST

MACHINE TYPE: ALL PROFILING AND CONTOURING MACHINES.

TEST CRITERIA:

MATERIAL: 7075-T6 ALUMINUM

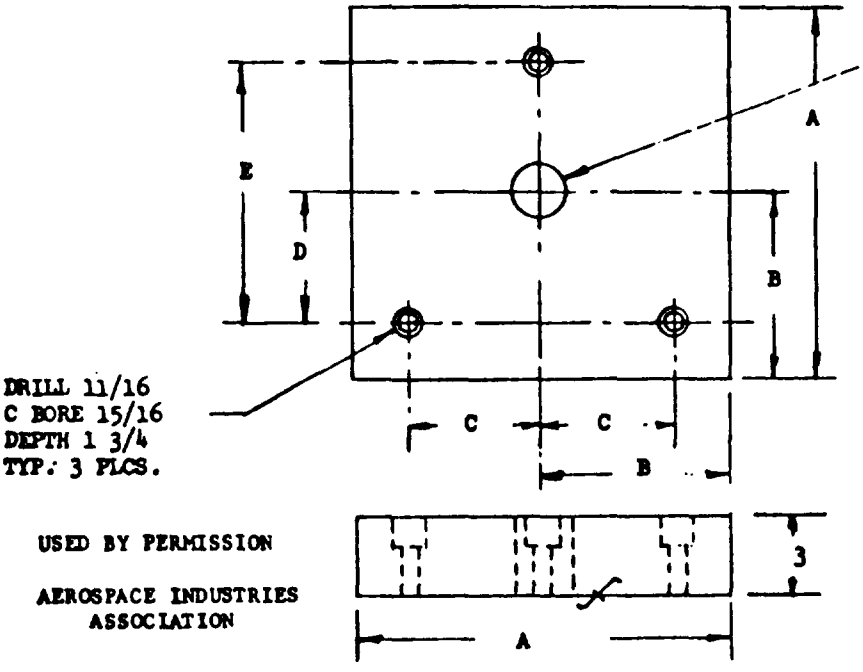
CUTTER: 2.000 INCH DIAMETER, 2 FLUTE END MILL, HSS, RIGHT HAND CUT, HIGH HELIX, 6 INCH MAXIMUM LENGTH TO MINIMIZE CUTTER DEFLECTION.

WIDTH AND DEPTH OF CUT: AS REQUIRED PER DIAGRAM - ROUGH AND FINISH CUTS ALLOWED.

WORK LOCATION: WORK PIECE TO BE CENTRALLY LOCATED ON THE TABLE WITH SIDES PARALLEL WITH MACHINE AXIS. PRE-BORED 2 INCH DIAMETER HOLE SHALL BE USED AS THE PROGRAMMED SET POINT.

BLANK PREPARATION:

DIM.	A	B	C	D	E
TYPE I	14	7	4 3/4	5	10.000 ± .010
TYPE II	8	4	USE MILL VISE FOR MAT. HOLD DOWN		



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NOTE: SURFACE AREA ½ INCH FROM CORNERS SHALL BE EXEMPT FROM EVALUATION. COMPLETELY REVISED (1)

NAS 979
SHEET 23

APPROVAL DATE APRIL 1969 REVISION (1) 15 Jan. 1969

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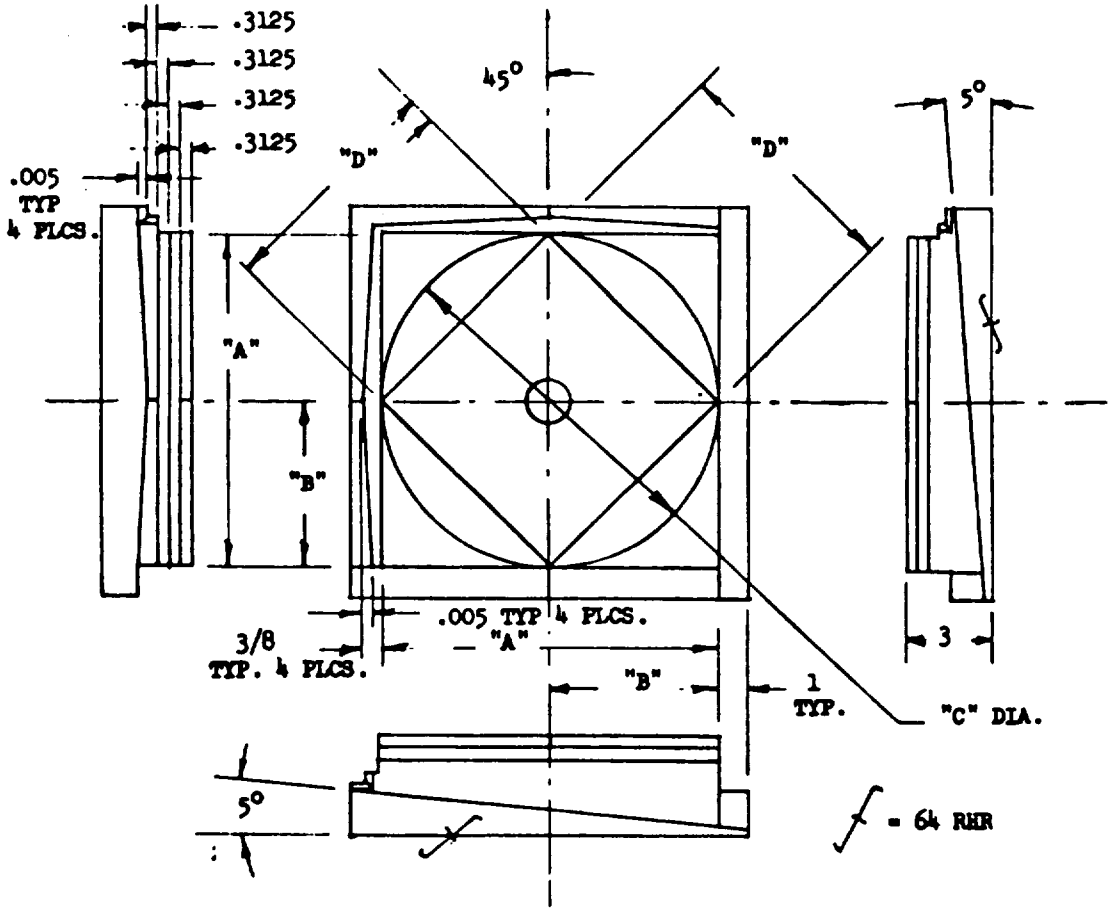
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4.3.3.5.1 (CONTINUED)

DIAGRAM:

NOTE: SURFACE AREA $\frac{1}{4}$ INCH FROM CORNERS SHALL BE EXEMPT FROM EVALUATION.



DIM.	"A"	"B"	"C"	"D"
TYPE I	12.0000	6.0000	12.0000	8.4853
TYPE II	6.0000	3.0000	6.0000	4.2426

VISUALLY INSPECT THE X, Y, AND Z AXES .005 RAMP AND TAPER CUTS FOR UNIFORMITY OF SERVO RESPONSE AND SLIDE STICTION.

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SHEET 24

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Flatness	0.002 inch TIR (flat surface of outside square)
Finish	35 rhr, all surfaces

4.7.3.4 Positioning accuracy and repeatability. The positioning accuracy and repeatability of the machine slides shall be evaluated in accordance with the National Machine Tool Builders' Association's Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools, Section 1 - Linear Definition. At least 10 random target points, spaced over the full travel of each axis shall be selected and approached from both directions.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The profilers covered by this specification are to be used in the shops and depots of activities in the Department of Defense.

6.2 Ordering data. Purchaser should exercise any desired option offered herein.

6.2.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2). If multiple spindles, specify distance between spindle centers (see 1.2).

c. First article, if required (see 3.1).

d. If the milling machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

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g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

i. Electrical system and input power, if different (see 3.4.12).

j. Motor enclosure, if different (see 3.4.12.1).

k. Computer numerical control (CNC), if different, specify requirements (see 3.4.14).

l. CNC power source, if different, specify requirements (see 3.4.14).

m. Input media, if different, specify requirements (see 3.4.14.1).

n. Tape reader spools, if different, specify requirements (see 3.4.14.1.1).

o. Thirty-two Bit Binary Cutter location (BCL), if required (see 3.4.14.2).

p. Additional control features, if different, specify requirements (see 3.4.14.3.8).

q. Auxiliary functions, if different, specify requirements (see 3.4.14.4.6).

r. Miscellaneous functions, if different, specify requirements (see 3.4.14.4.6).

s. Data display, if different, specify requirements (see 3.4.14.5.1).

t. Operator control panel, if different, specify requirements (see 3.4.14.5.14).

u. Part program storage, if different, specify requirements (see 3.4.14.6.1).

v. Standby battery power, if different, specify requirements (see 3.4.14.6.2).

w. Canned cycles, if different, specify requirement (see 3.4.14.6.8).

x. Operator portable or pendant control, if required, specify requirement (see 3.4.14.6.9).

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y. Postprocessor, specify part program language, host computer (including model number, serial number and any modifications or enhancements to operational features), (see 3.4.14.6.10).

z. Additional data input media, when required, specify requirements (see 3.4.14.6.11).

aa. Reader recorder, when required, specify requirements (see 3.4.14.6.12).

bb. Portable tape punch unit, when required, specify requirement (see 3.4.14.6.12).

cc. Tracer machining capability, when required, specify requirement (see 3.4.14.6.13).

dd. Digitizing capability, when required, specify requirements (see 3.4.14.6.14).

ee. Standard equipment, if different, specify requirement (see 3.8).

ff. Optional equipment, specify requirement (see 3.9).

gg. Additional performance requirement, specify requirement (see 3.10).

hh. Lubrication chart, or plate, if different, specify requirement (see 3.11.1).

ii. Nameplate, if different, specify requirement (see 3.11.2).

jj. Hour meter, if different, specify requirement (see 3.12).

kk. Inspection conditions, if different, specify requirements (see 4.3).

ll. First article inspection, if different, specify requirement (see 4.4 and 6.3).

mm. Quality conformance inspection, if different, specify requirement (see 4.5).

nn. Tests, if different, specify requirement (see 4.7).

oo. Preservation, packing and marking, specify required level (see 5.1).

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6.2.2 Contact data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports) should be specified on a DD Form 1423, Contact Data Requirements Lists, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the milling machine itself. It does not require compliance with those OSHA requirements that concern "the milling machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual milling machine in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new milling machine into its future operating environment. The milling machine shall be equipped with all point-of-operation guarding supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.f.

6.5 Subject term (key word) listing.

Computer numerical controlled
Machine tools
Milling machine
Profiler, vertical
Spindle, single, multiple
Tracer controlled

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Type I, class 1								
	Type II, classes 2, 3 and 4			Type III, classes 2, 3, 4, 5, and 6					
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9
Table work surface, width x length, inches	28 x 72	28 x 119	28 x 143	40 x 72	40 x 119	40 x 143	30 x 119	40 x 119	40 x 143
T-slots, number and size	5, 13/16	5, 13/16	5, 13/16	7, 13/16	7, 13/16	7, 13/16	5, 13/16	7,13/16	7, 13/16
Travel range, inches									
Longitudinal (X)	60	96	120	60	96	120	60	120	144
Cross (Y)	30	30	30	40	40	40	30	30	30
Vertical (Z)	14	14	14	14	14	14	18	24	24
Quill adjustment (manual) inch	1	1	1	1	1	1	2	2	2
Feed rates, IPM, ±10 percent									
X	0.5-100	0.5-100	0.5-100	0.5-100	0.5-100	0.5-100	0-120	0-120	0-150
Y	0.5-100	0.5-100	0.5-100	0.5-100	0.5-100	0.5-100	0-120	0-150	0-150
Z	0.5-50	0.5-50	0.5-50	0.5-50	0.5-50	0.5-50	0-60	0-60	0-60
Rapid traverse, IPM, ±10 percent									
X	100	100	100	100	100	100	120	180	180
Y	100	100	100	100	100	100	120	180	180
Z	50	50	500	50	50	50	60	60	60
Spindle nose taper, Natl. Standard	50	50	500	50	50	50	50	50	50
Spindle speeds, number of	24	24	24	24	24	24	Infinite	Infinite	Infinite
Spindle speeds range, RPM standard ±10 percent	18-1800	18-1800	18-1800	18-1800	18-1800	18-1800	18-1800	18-1800	18-1800
Spindle drive motor hp minimum (std.)									
Option A, minimum	20	20	20	25	25	25	20	20	20
Option B, minimum	25	25	25	30	30	30	—	—	—
Spindle drive motor shared/multi-spindle	40	40	40	49	40	40	—	—	—
HP/RPM/spindle, type II machines	yes	yes	yes	yes	yes	yes	no	no	no
HP/RPM/spindle, type I and III machines	1/10	1/10	1/10	1/10	1/10	1/10	—	—	—
	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4

NOTE: Machines sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE II. Alignment accuracies.

Test	Tolerances
Spindle(s) periphery runout	0.0004" TIR
Spindle(s) face runout	0.0003" TIR
Spindle(s) internal taper runout	
1 inch from spindle(s) nose	0.0004" TIR
12 inches from spindle(s) nose	0.001" TIR
Longitudinal rise and fall of table	0.0008" in any 12-inch span. Total not to exceed ±0.0005" + 0.00015" per each 12 inches of travel.
Cross slide ways parallel with table	0.0017" full travel
Cross movement square with T-slots	0.0005" in any 12-inch span. Total not to exceed ±0.0005" + 0.00015" per each 12 inches of travel.
Spindle(s) square with table, center of rotation:	
a. Single spindle	±0.0005" per 12-inch sweep
	0° to 180°
	90° to 270°
b. Multiple spindle	±0.001" per 12-inch sweep
	0° to 180°
	90° to 270°
Spindle(s) vertical travel square with table	0.0015" in any 12-inch span. Not to exceed .002" over all.
T-slots parallel to table travel	0.001" in any 12-inch span. Not to exceed .002" over all.
Centerlines of spindles parallel with respect to each other	±0.002"

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TABLE III. Diskette addressing.

Sector address	1/ Track	1/ Sector number	Byte	Use
0 - 25	0	All	All	Unused
20 - 29	1	1 - 4	0 - 1	File start sector (binary)
			2 - 31	30-character ASC II defined by user program name to be
			32 - 63	Same as 2 - 31 above for next part program (file)
			64 - 95	Same as 32 - 63 above for next part program
			96 - 127	Same as 32 - 63 above for next part program
		5 - 26	All	APT CL records for NC part program
30 - 1897	2 - 73	All	All	APT CL records for NC part
1898 - 1975	74 - 76	All	All	Unused

NOTE 1/ This numbering corresponds to IBM 3740 physical format.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR
DLA - GS
Navy - YD
Air Force - 84

User activities:
Army - SM, AR
Navy - AS, OS

Preparing activity:
DLA - IP

Project (3417-0161)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80212A	Milling Machines, Vertical Single and Multiple Spindle, Computer Numerically Controlled (CNC) and Tracer

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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